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**CPT111 Java Programming**

**Week 6 Lab Exercises**

# **Introduction to Objects**

# Coding and Submission

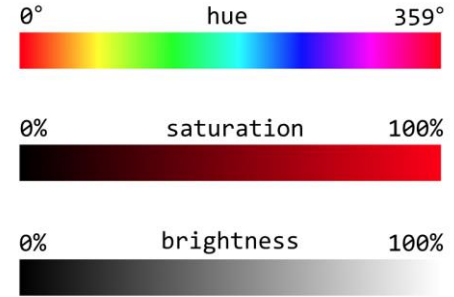
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- Coding in your NetBeans
  - Start with the skeleton code given in the course LMO
  - Add your own test cases
- Submitting into Learning Mall Quiz
  - Do not submit the whole class
  - Only submit the constructor or the method
    - read carefully the instructions
  - You can submit your own private helper method
    - but do not add another public methods

## Lab Exercise #6 ColorHSB

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- Complete a class ColorHSB that represents a color in hue – saturation – brightness (HSB) format
  - A skeleton file ColorHSB.java is given
- A color in HSB format is composed of three components:
  - Hue is an integer between 0 and 359
    - It represents a pure color on the color wheel in degrees ( $^{\circ}$ ), with  $0^{\circ}$  for red,  $120^{\circ}$  for green, and  $240^{\circ}$  for blue
  - Saturation is an integer between 0 and 100
    - It represents the purity of the hue in percentage (%)
  - Brightness is an integer between 0 and 100
    - It represents the percentage (%) of white that is mixed with the hue



## Lab Exercise #6.1 ColorHSB Constructor

- Complete the constructor of the class ColorHSB
- It takes three arguments: h, s, and b
  - and creates a new ColorHSB object with hue h, saturation s, and brightness b
- Test cases:

```
ColorHSB green = new ColorHSB(100, 100, 50);
```

`System.out.println(green);` → (100, 100, 50)

## Lab Exercise #6.2 ColorHSB toString

- Complete the method toString of the class ColorHSB
- It takes no argument
  - and return a string composed of the integers for hue, saturation, and brightness (in that order), separated by commas, and enclosed in parentheses
- Test cases:

```
ColorHSB green = new ColorHSB(100, 100, 50);
```

`System.out.println(green);` → (100, 100, 50)

## Lab Exercise #6.3 ColorHSB isGrayscale

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- Complete the method isGrayscale of the class ColorHSB
- A color in HSB format is a shade of gray if either its saturation or brightness component is 0% (or both)
  - return true if it is, and false otherwise
- Test cases:

ColorHSB orange = new ColorHSB(25, 100, 100);

ColorHSB gray = new ColorHSB(0, 0, 50);

System.out.println(orange.isGrayscale());                      → false

System.out.println(gray.isGrayscale());                        → true

## Lab Exercise #6.4 ColorHSB squareDist

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- Complete the method squareDist of the class ColorHSB that returns the squared distance between two colors
- The squared distance between two colors  $(h_1, s_1, b_1)$  and  $(h_2, s_2, b_2)$  is defined to be

$$\min\{(h_1 - h_2)^2, (360 - |h_1 - h_2|)^2\} + (s_1 - s_2)^2 + (b_1 - b_2)^2$$

- Test cases:

ColorHSB green = new ColorHSB(100, 100, 50);

ColorHSB orange = new ColorHSB(25, 100, 100);

ColorHSB gray = new ColorHSB(0, 0, 50);

int distGreenOrange = green.squareDist(orange);

System.out.println(distGreenOrange);                      → 8125

System.out.println(gray.squareDist(orange))              → 13125



# Thank you for your attention!

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- This is the end of Week 6 Lab Exercise